

Name: _____ Partner: _____ Period: _____

Context: Transformations of Sinusoidal Functions

The general form $f(\theta) = a \sin(b(\theta + c)) + d$ encodes four independent features of a sinusoidal function. In this activity your group will decode these parameters from equations, connect them to transformations, and reason about how changes to each parameter affect the function.

Tier R — Reinforce (Key)

Function	a	b	Amplitude	Period	Midline
$y = 3 \sin(\theta) + 1$	3	1	3	2π	$y = 1$
$y = -\sin(2\theta)$	-1	2	1	π	$y = 0$
$y = \frac{1}{2} \sin(\pi\theta) - 3$	$\frac{1}{2}$	π	$\frac{1}{2}$	2	$y = -3$
$y = 5 \cos(\theta) + 2$	5	1	5	2π	$y = 2$
$y = -4 \cos(3\theta) - 1$	-4	3	4	$2\pi/3$	$y = -1$

- Maximum = **4** Minimum = **-2**
- The negative sign reflects the graph over its midline ($y = 0$), so the graph starts at 0, goes down first (to -1) instead of up.**
- Maximum = $(-3) + (1/2) = \mathbf{-2.5}$ Minimum = $(-3) - (1/2) = \mathbf{-3.5}$
- $f(\theta) = \mathbf{6 \sin(2\theta) - 2}$ (amplitude 6: $a = 6$; period π : $b = 2$; midline -2: $d = -2$; no phase shift: $c = 0$)

Tier A — Apply (Key)

Function 1: $f(\theta) = 2 \sin\left(3\left(\theta + \frac{\pi}{6}\right)\right) - 1$

Amplitude: **2** Period: $2\pi/3$ Phase shift: $-\pi/6$ (**left $\pi/6$**) Midline: $y = -1$

Function 2: $g(\theta) = -3 \cos(2\theta + \pi) + 4$

Factor: $2\theta + \pi = 2(\theta + \pi/2)$

Amplitude: **3** Period: π Phase shift: $-\pi/2$ (**left $\pi/2$**) Midline: $y = 4$

Function 3: $h(\theta) = 4 \sin\left(\frac{1}{2}\theta - \frac{\pi}{4}\right)$

Factor: $\frac{1}{2}\theta - \frac{\pi}{4} = \frac{1}{2}(\theta - \pi/2)$

Amplitude: **4** Period: 4π Phase shift: $\pi/2$ (**right**) Midline: $y = 0$

Function 4: $k(\theta) = \cos\left(2\theta - \frac{\pi}{3}\right) + 2$

Factor: $2(\theta - \pi/6)$

Amplitude: **1** Period: π Phase shift: $\pi/6$ (**right**) Midline: $y = 2$

Discussion: Yes. Use $\cos \theta = \sin(\theta + \pi/2)$. For Function 2: $-3 \cos(2(\theta + \pi/2)) + 4 = -3 \sin(2(\theta + \pi/2) + \pi/2) + 4 = -3 \sin(2\theta + \pi + \pi/2) + 4$, or equivalently shift the cosine form by an additional $\pi/2$ to convert to sine. Each cosine function requires an extra phase shift of $\pi/2$ to become a sine function.

Tier E — Extend (Key)

Function A: $p(\theta) = -3 \cos(2\theta - \pi/3) + 1$

Step 1: $2(\theta - \pi/6)$

Step 2: amplitude **3**, period π , phase shift $\pi/6$ **right**, midline $y = 1$, reflected **yes**.

Step 3: $p(\theta) = -3 \sin\left(2\left(\theta - \pi/6 + \pi/4\right)\right) + 1 = -3 \sin\left(2\left(\theta - \pi/12\right)\right) + 1$

Teacher Note

Step 3 detail: $\cos \theta = \sin(\theta + \pi/2)$, so $\cos(2(\theta - \pi/6)) = \sin(2(\theta - \pi/6) + \pi/2) = \sin(2\theta - \pi/3 + \pi/2) = \sin(2\theta + \pi/6) = \sin(2(\theta + \pi/12))$. Therefore $p(\theta) = -3 \sin(2(\theta + \pi/12)) + 1$, phase shift $= -\pi/12$ (left).

Step 4: **Start with $y = \sin \theta$. Apply horizontal compression by factor $1/2$ (period becomes π). Shift left $\pi/12$. Stretch vertically by factor 3. Reflect over the midline. Shift up 1 unit.**

Function B: $q(\theta) = 5 \sin(3\theta + \pi/2) - 2$

Step 1: $3(\theta + \pi/6)$

Step 2: amplitude **5**, period $2\pi/3$, phase shift $-\pi/6$ (**left**), midline $y = -2$.

Step 3: $q(\theta) = 5 \cos(3\theta) - 2$ (since $\sin(\theta + \pi/2) = \cos \theta$)

Step 4: **Start with $y = \sin \theta$. Compress horizontally by factor $1/3$ (period $= 2\pi/3$). Shift left $\pi/6$. Stretch vertically by factor 5. Shift down 2.**

Challenge: No. Two functions with the same amplitude and period can still differ in phase shift and midline. Four parameters — a , b , c , d — are required to uniquely determine a sinusoidal function (up to whether sine or cosine is used as the parent).